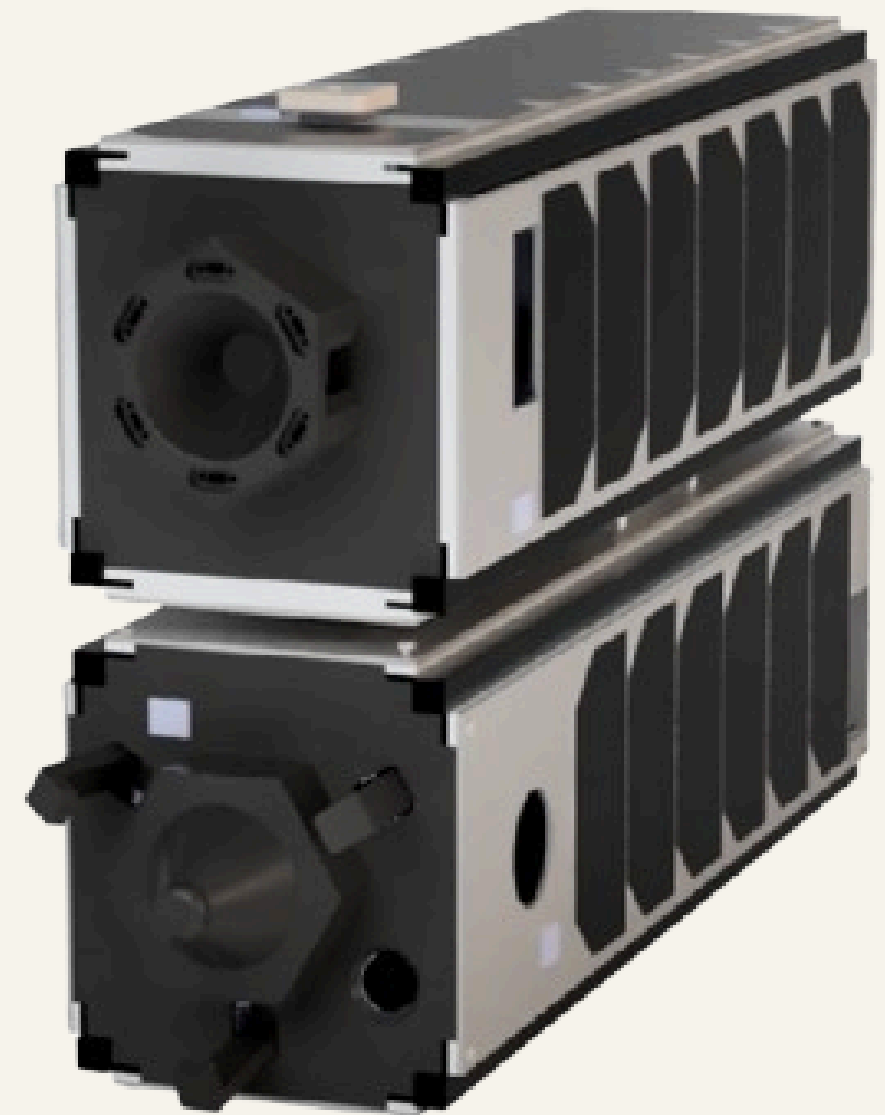


ME701

23B2120 | Aditya Nimbalkar

Docking Adapter

Latch Optimization



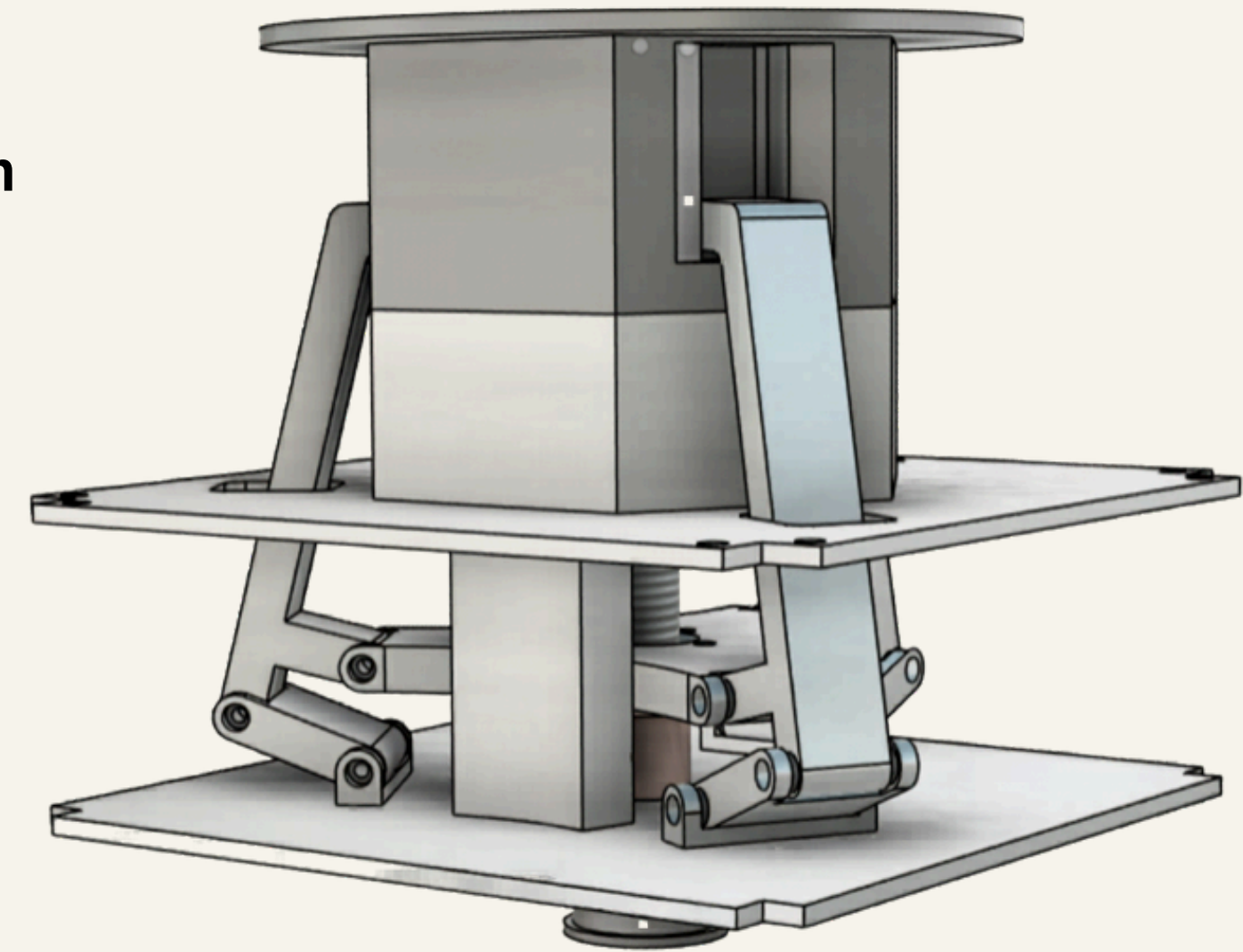
OVERVIEW

- Problem Statement - 03
- Notation - 04
- Constraints - 06
- Objective Function - 07
- Optimizer - 11
- Results - 12

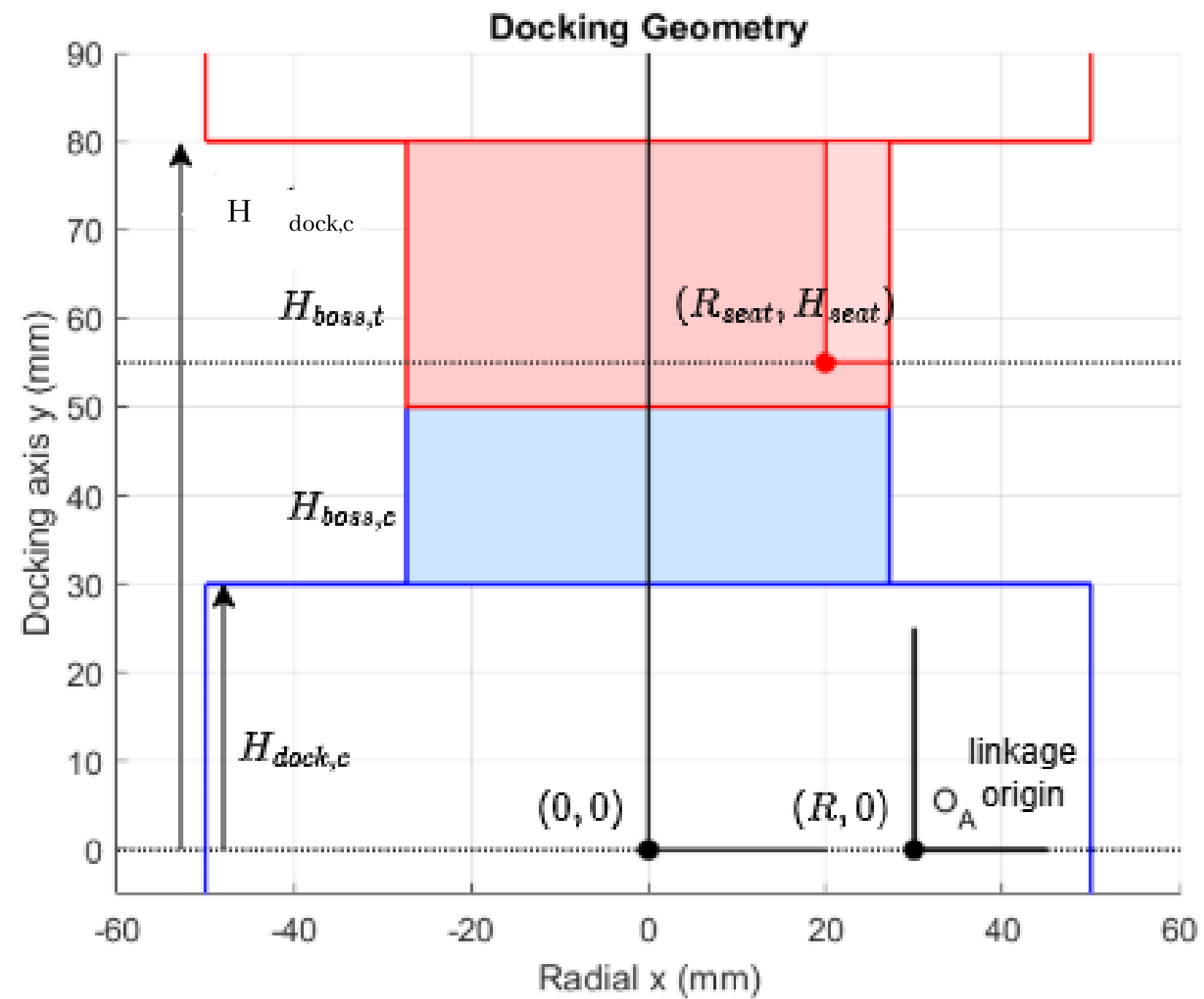


PROBLEM STATEMENT

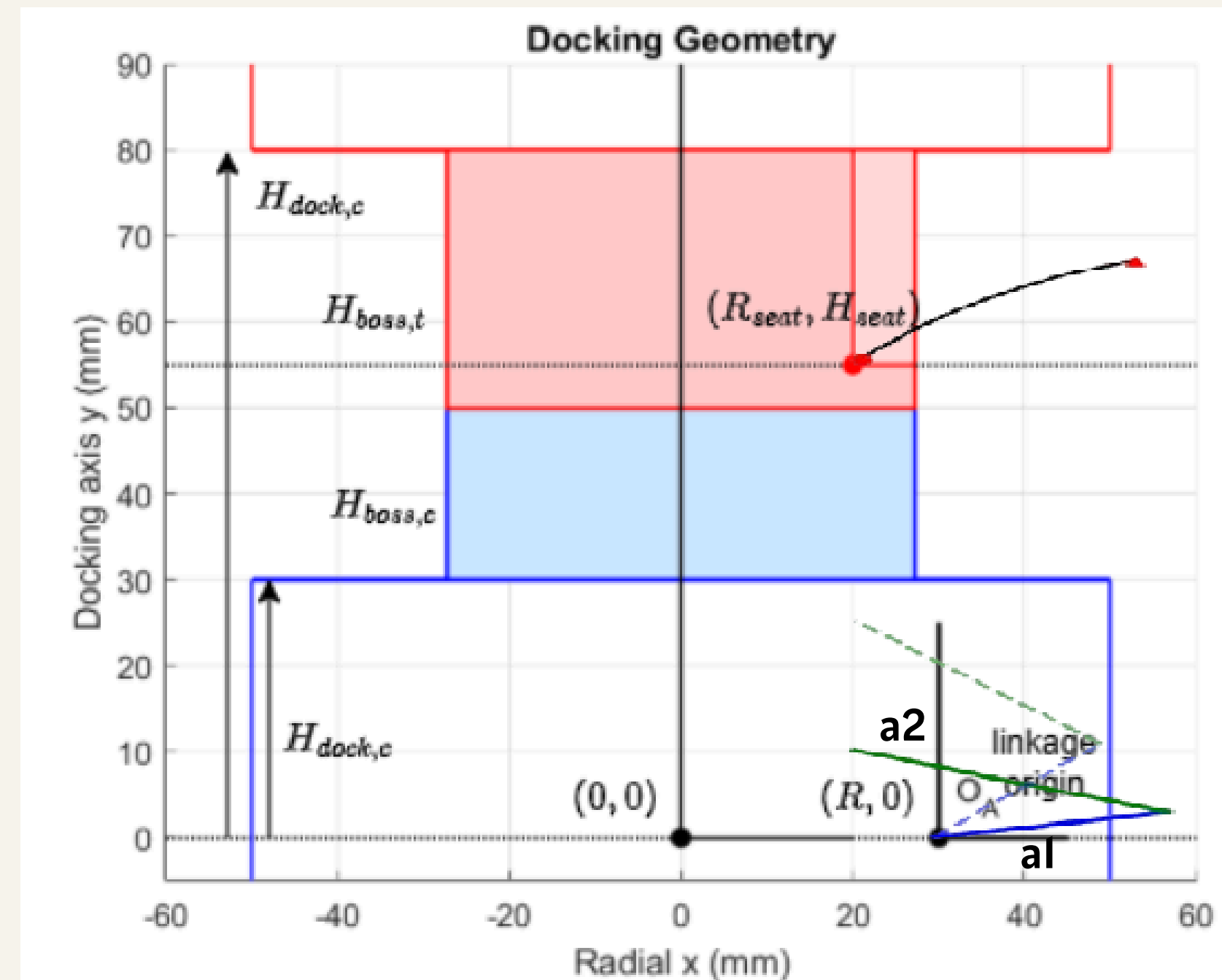
- The Docking Adapter Unit (DAU) uses a three-latch mechanism, each implemented as a planar RRRP linkage actuated by a linear slider.
- The slider motion produces a coupled radial and axial motion of the latch tip:
 - Radial motion allows the latch to enter the recess of the female docking interface
 - Axial motion pulls the two docking ports together and generates preload DAU-work-updates
- The geometry of this linkage determines the trajectory of the latch tip and the force transmission characteristics of the mechanism.
- A feasible linkage geometry has already been obtained through preliminary synthesis, but it is not optimal.



NOTATION



NOTATION



CONSTRAINTS

Latch Tip

Radial Clearance: $x_{\text{open}} \geq R_{\text{hex}}$

Interference Mitigation: $x \geq R_{\text{seat}}$

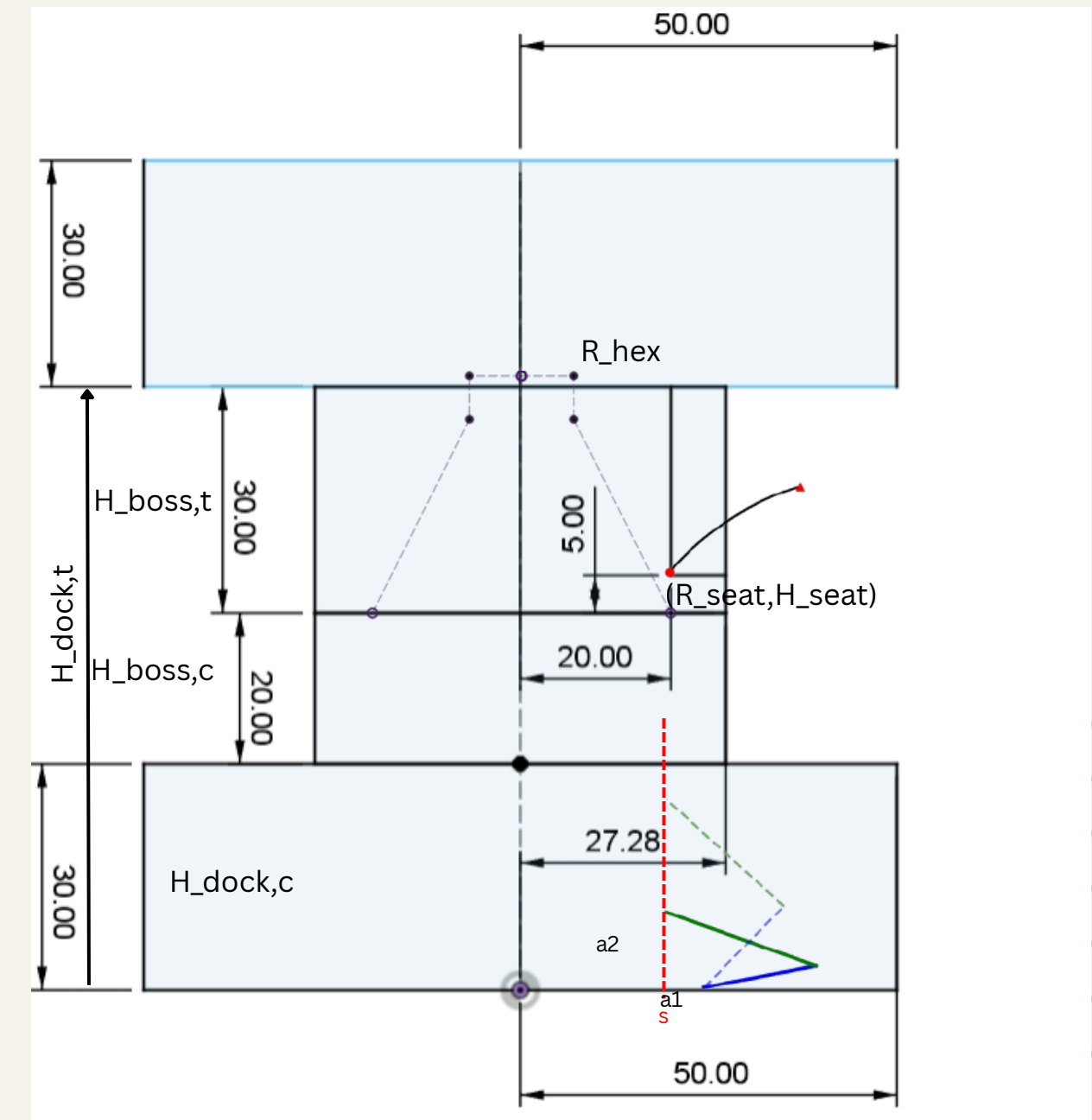
Vertical Bounds: $H_{\text{seat}} \leq y_{\text{open}} \leq H_{\text{dock,t}}$

Packaging

Maximum Allowable Radial Footprint: $a1 \leq R_{\text{max}} - R$

Hardware Interference: $R - S \geq 5\text{mm}$

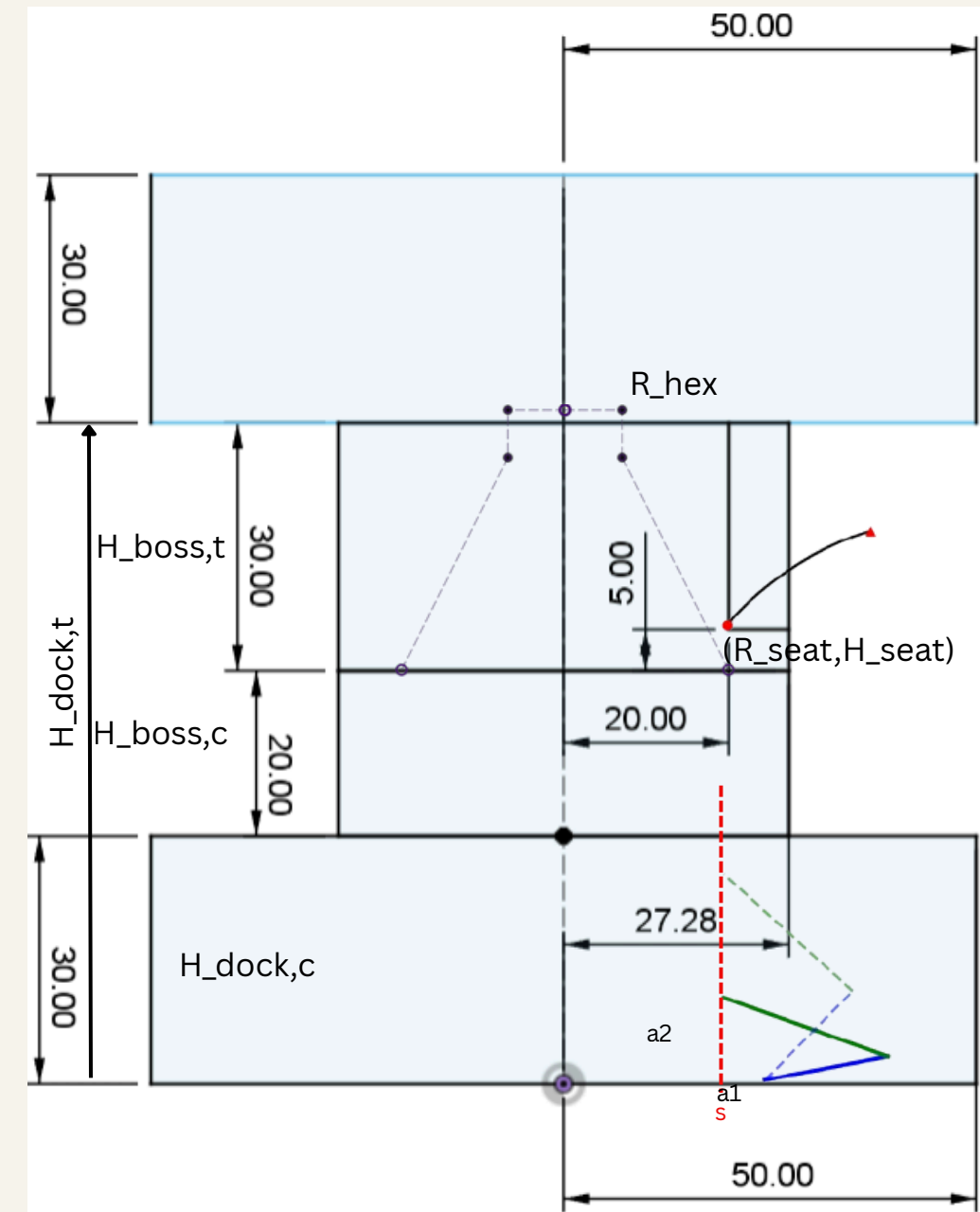
"Upward Bending Right Elbow"



OBJECTIVE FUNCTION

PERFORMANCE TARGETS

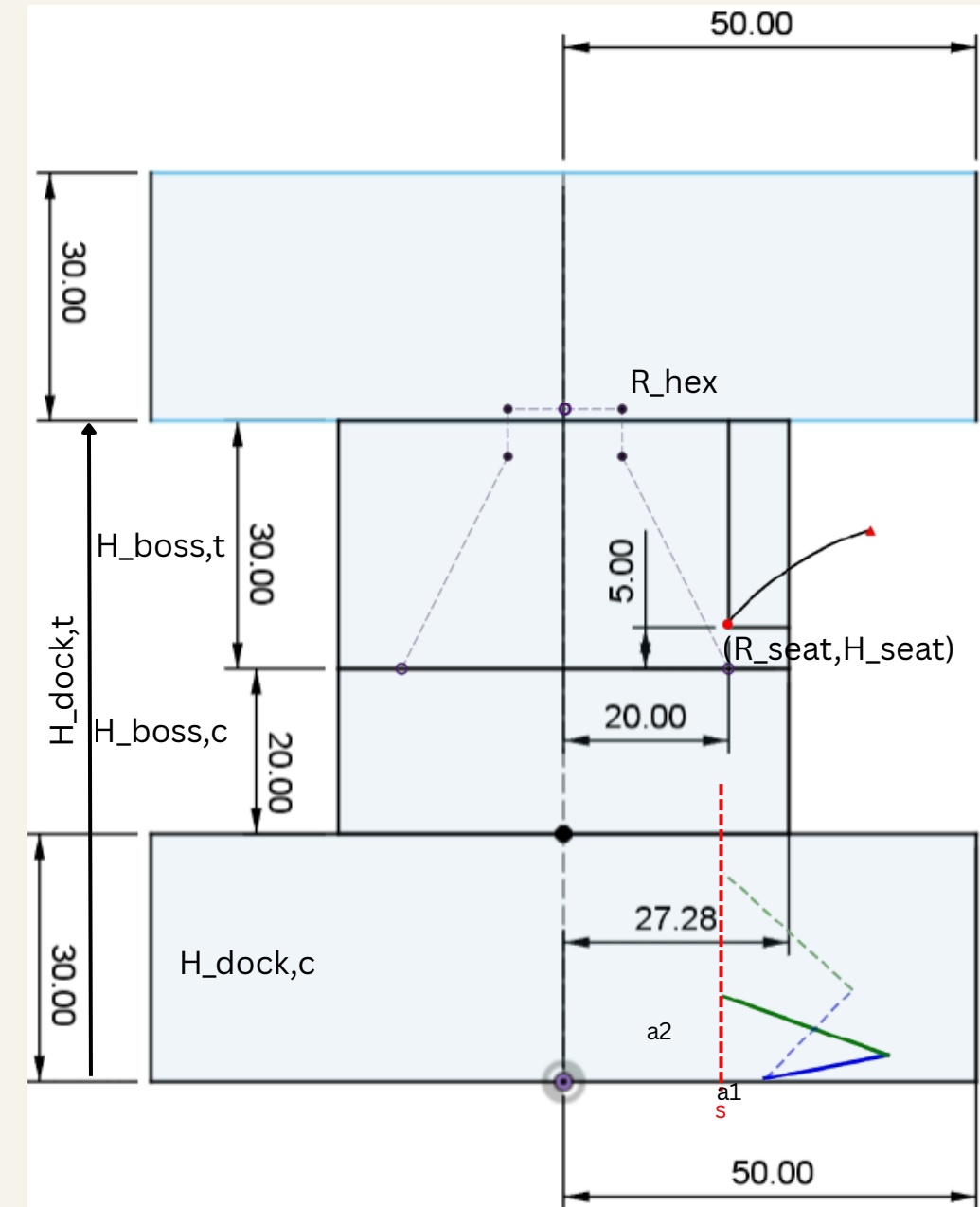
- **Closed Position Accuracy (Weight: 20):** Heavily penalizes any deviation from the precise target seat coordinates (R_{seat} , H_{seat}) at the end of the stroke .
- **Capture Capability (Weight: 15):** Rewards designs that open wider than the required R_{hex} boundary, maximizing lateral capture margin .
- **Axial Tolerance (Weight: 10):** Rewards trajectories that maintain high clearance above the seat as the tip crosses the entry threshold, maximizing vertical misalignment tolerance .



OBJECTIVE FUNCTION

KINEMATIC METRICS

- **Monotonic Motion (Weight: 15):** Enforces strict inward and downward motion; heavily penalizes any trajectory that zig-zags or reverses direction to prevent jamming .
- **Force Transmission (Weight: 8):** Rewards high mechanical advantage. By favoring lower tip velocities projected to downward motion, the mechanism guarantees higher pulling force when latching .
- **Smoothness (Weight: 2):** Minimizes trajectory acceleration (second derivative) to ensure smooth, predictable movement.



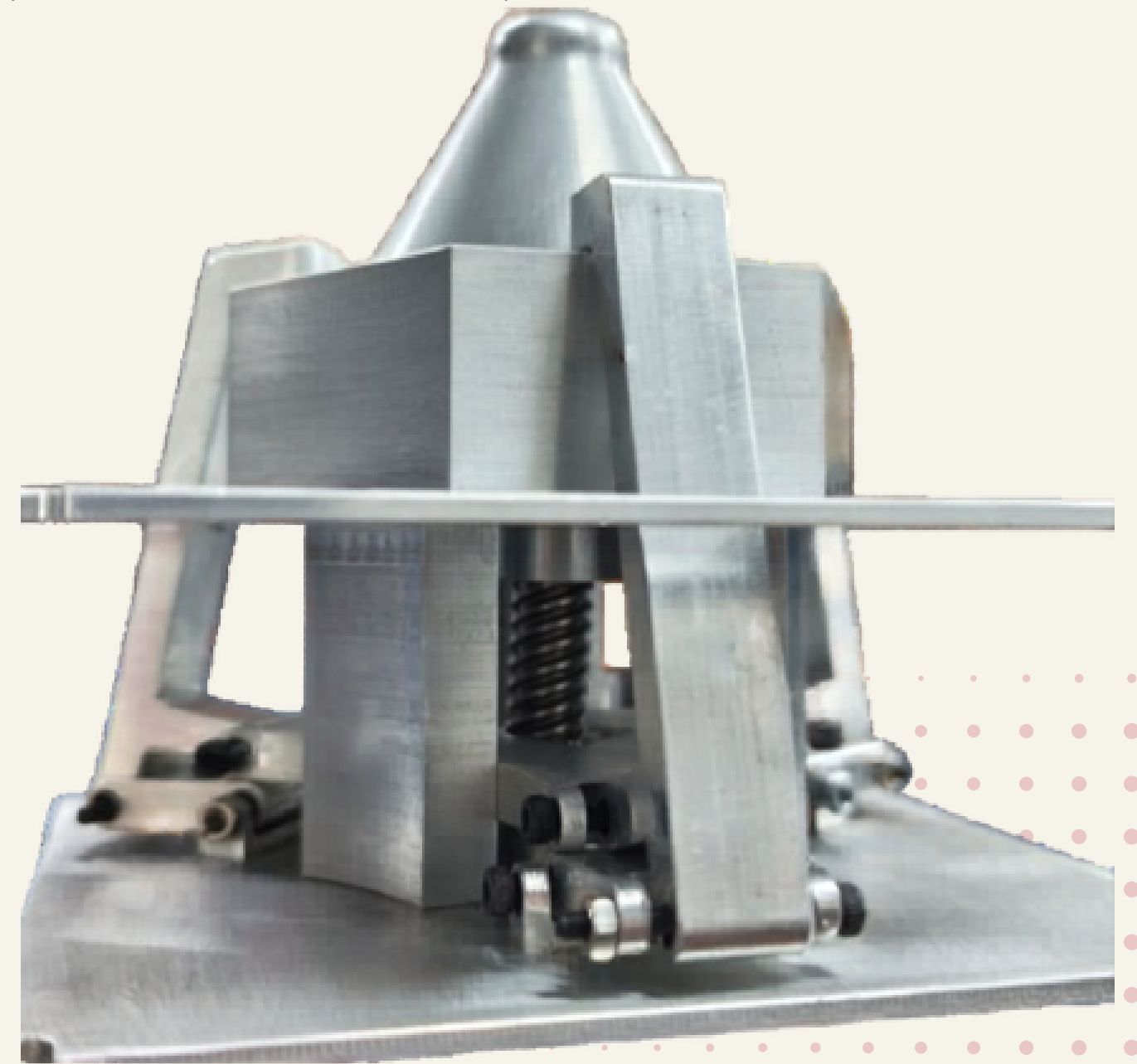
Gradient-Friendly Walls (Weight: 50): Hard geometric and physical constraints are converted into smooth quadratic penalties (e.g., $\max(0, \text{"violation"})^2$). This prevents the optimizer from getting "blinded" by abrupt mathematical cliffs .

- [illegible]

OBJECTIVE FUNCTION

OVERALL

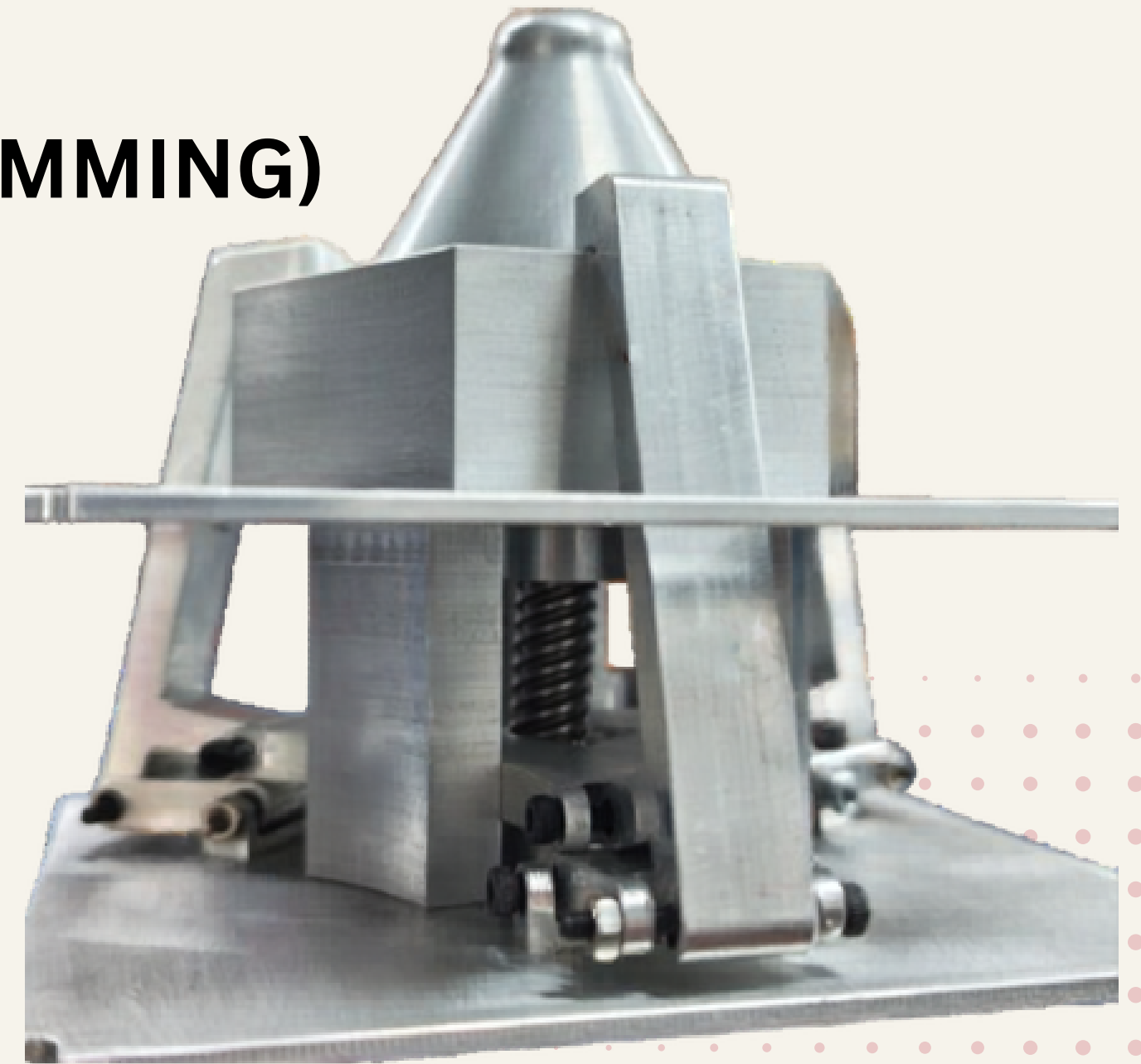
$$\begin{aligned} &20 * J_{\text{closed}} + \\ &20 * J_{\text{capture}} + \\ &10 * J_{\text{axial}} + \\ &15 * J_{\text{mono}} + \\ &2 * J_{\text{smooth}} + \\ &8 * J_{\text{force}} + \\ &50 * \text{Penalty} \end{aligned}$$



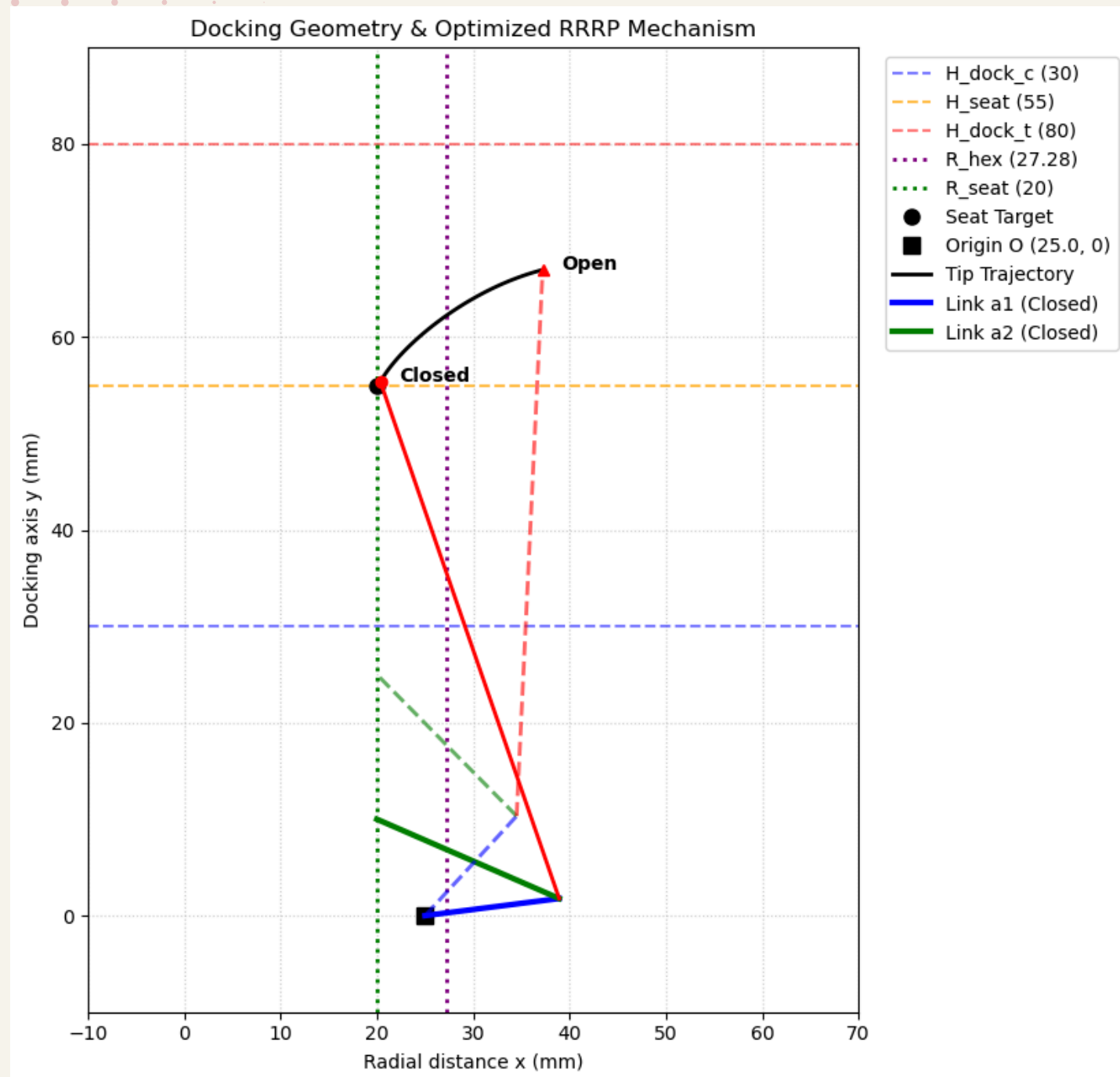
OPTIMIZER

SLSQP (SEQUENTIAL LEAST SQUARES PROGRAMMING)

- Gradient-Based
- SciPy Library
- Allows bounding variables
- Great for Constraint Non-Linear Optimization Problems



RESULTS



OPTIMIZED MECHANISM PARAMETERS

Link 1 (a1) : 14.039 mm

Link 2 (a2) : 20.635 mm

Origin (R) : 25.000 mm

Slider Pos (s) : 20.000 mm

Coupler offset (u,v) : (38.372, -41.745) mm

KEY PERFORMANCE METRICS

Seat Target Error : 0.510 mm (dx: 0.366, dy: 0.355)

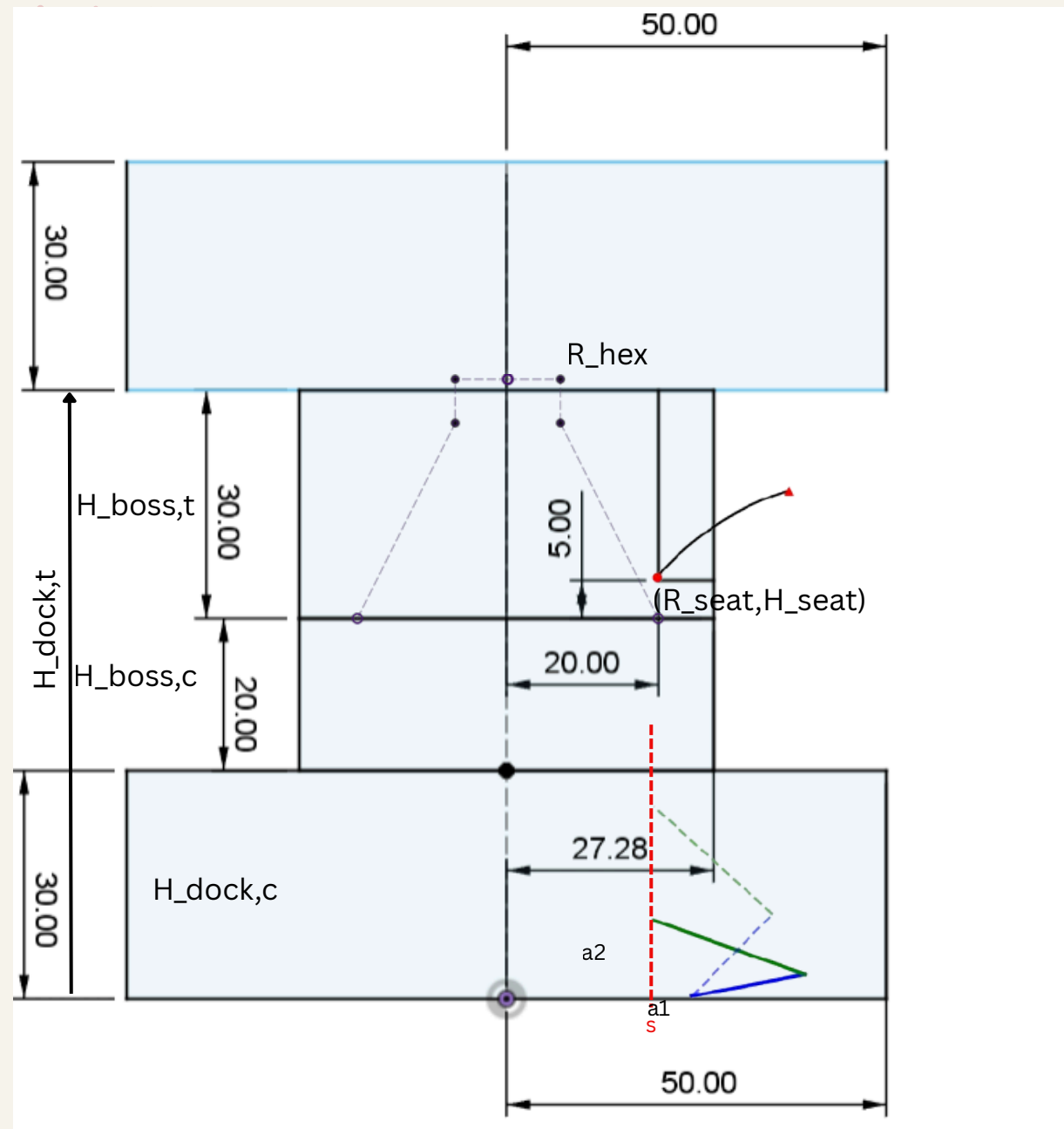
Lateral Capture Margin: 9.931 mm (Positive = correctly outside R_hex)

Open Tip Height : 66.964 mm (Constraint: < 80.0)

Axial Tolerance : 7.268 mm (Height margin above seat when passing R_hex)

Max Reach Required : 25.495 mm

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The background features three vertical stripes on the left: a wide pink stripe, a medium blue stripe, and a narrow beige stripe. The right side of the image is a light beige background with two rectangular areas of small, light pink dots in the top right and bottom right corners.

THANK YOU